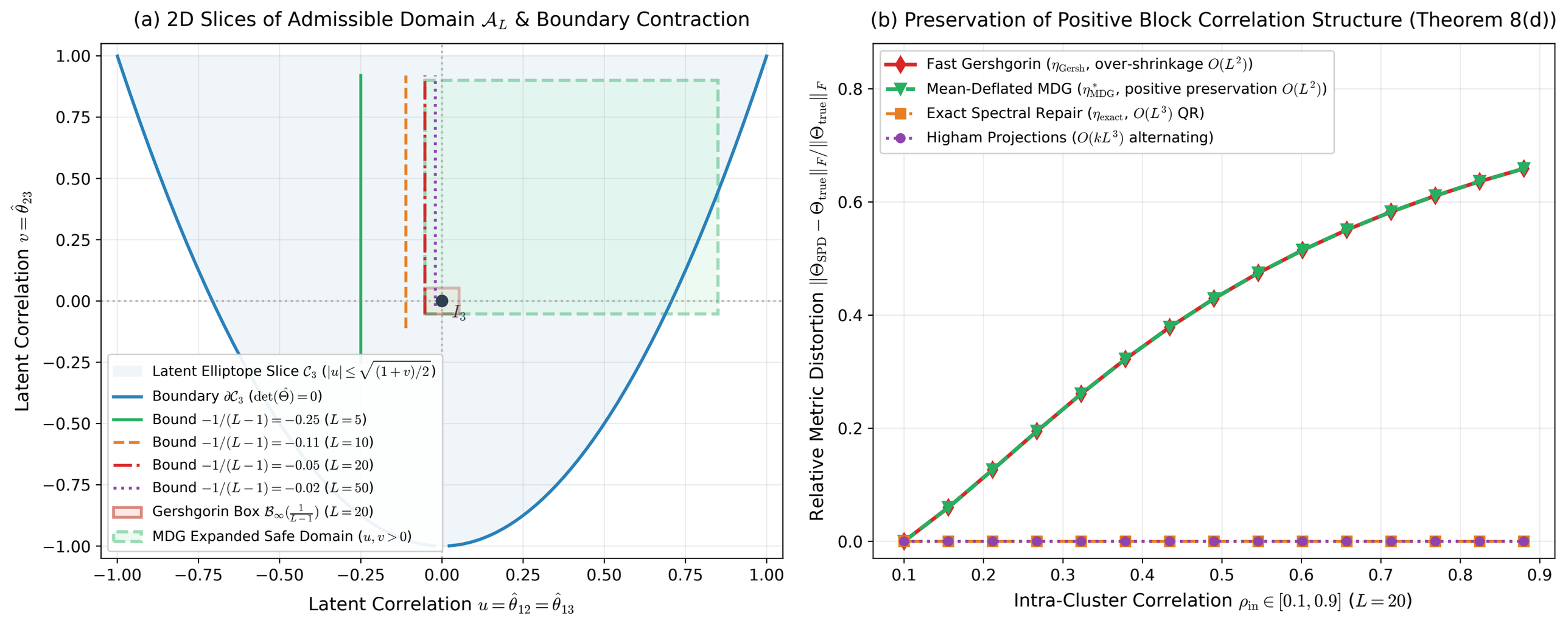
(a) 2D Slices of Admissible Domain $\mathcal{A}_L$ & Boundary Contraction
Latent Correlation $v = \hat{\theta}_{23}$
Latent Correlation $u = \hat{\theta}_{12} = \hat{\theta}_{13}$
$I_3$
Latent Elliptope Slice $\mathcal{C}_3$ ($|u| \leq \sqrt{(1+v)/2}$)
Boundary $\partial\mathcal{C}_3$ ($\det(\hat{\Theta}) = 0$)
Bound $-1/(L-1) = -0.25$ ($L = 5$)
Bound $-1/(L-1) = -0.11$ ($L = 10$)
Bound $-1/(L-1) = -0.05$ ($L = 20$)
Bound $-1/(L-1) = -0.02$ ($L = 50$)
Gershgorin Box $\mathcal{B}_\infty(\frac{1}{L-1})$ ($L = 20$)
MDG Expanded Safe Domain ($u, v > 0$)
(b) Preservation of Positive Block Correlation Structure (Theorem 8(d))
Relative Metric Distortion $\|\Theta_{\text{SPD}} - \Theta_{\text{true}}\|_F / \|\Theta_{\text{true}}\|_F$
Intra-Cluster Correlation $\rho_{\text{in}} \in [0.1, 0.9]$ ($L = 20$)
Fast Gershgorin ($\eta_{\text{Gersh}}$, over-shrinkage $O(L^2)$)
Mean-Deflated MDG ($\eta^*_{\text{MDG}}$, positive preservation $O(L^2)$)
Exact Spectral Repair ($\eta_{\text{exact}}$, $O(L^3)$ QR)
Higham Projections ($O(kL^3)$ alternating)